

AYAAN AKHTER

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Education

VIT Bhopal University — Bhopal, India

Bachelor of Technology in Computer Science Engineering, Specialization in AI/ML

Expected June 2027

Technical Skills

Programming Languages: Python, C++, SQL

AI & Machine Learning: Deep Learning, Computer Vision, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG)

Developer Tools & Frameworks: PyTorch, Git, GitHub, REST APIs

Core CS & Software Engineering: Operating Systems, Computer Networks, Database Management Systems (DBMS), System Design Fundamentals

Experience

Undergraduate Research Assistant

Aug 2024 – Present

- Conducted applied research in deep learning and computer vision architectures, engineering automated evaluation frameworks to benchmark model performance across 12+ experimental pipelines.
- Performed system modeling and algorithmic optimization, achieving a 30% reduction in inference latency and improving overall computational efficiency.
- Maintained structured technical documentation and adhered to clean-code practices across complex experimental workflows.

Project Collaborator [Certificate] – Collaborative AI Developer

Dec 2025 – Mar 2026

- Collaborated with a global team of engineers to build and deploy end-to-end machine learning systems addressing real-world problem statements.
- Engineered modular data pipelines and integrated AI models with automated performance testing and responsible AI metrics.
- Participated in agile peer reviews, sprint planning, and version control workflows to deliver production-grade code.

Projects

Hybrid Vision Transformer Glaucoma Classifier [Repository]

Python, PyTorch, Swin Transformer, ConvNeXt, Model Calibration

- Architected a dual-backbone deep learning framework (Swin-L + ConvNeXt-L) with CBAM attention, achieving 95.15% peak validation accuracy across 5-fold cross-validation.
- Reduced validation loss by 64% (to 0.1637) within 7 epochs using class-weighted Focal Loss and an auxiliary detection head to minimize false negatives.
- Evaluated Expected Calibration Error (ECE) across 15 confidence bins and Brier Scores to ensure clinically reliable, risk-calibrated output probabilities.

CodeMode Challenge Project [Repository]

Python, Model Fine-Tuning, Dataset Creation

- Designed fine-tuned embeddings and curated syntax-aware datasets for agentic codebases, improving semantic retrieval throughput by 28%.
- Implemented automated unit testing and validation frameworks, reducing edge-case latency by 35% under heavy system workloads.

Leadership & Achievements

- Finalist, VIT Ideathon — Conceptualized and presented scalable technology solutions to real-world operational challenges.
- Table Topics Master & Member, Toastmasters Club — Led public speaking sessions, demonstrating strong impromptu communication, structured thinking, and team leadership.